The Relay — Research Paper Assistant

1. What I built and why

A Python assistant retrieves passages from three specified arXiv papers and uses Groq (openai/gpt-oss-20b) for streamed, cited answers. It refuses unsupported questions. The corpus has 384 chunks (220 words, 40 overlap): 202 Moshi, 142 Dynamic Chunking and 40 Attention. The pipeline uses TF-IDF top three.

2. The real service and my stand-in

Groq is the hosted service. My configurable asynchronous stand-in simulates latency, streaming, rate limits, failures, timeouts and overload using canned text. SDK retries are disabled; I implemented two retries after the initial attempt, using Retry-After or exponential backoff. Partial streams are not retried.

Table 1 — Real service limits

Limit	Finding	Confidence
RPM, RPD, TPD, concurrency	Not measured	—
TPM	8,000	High; actual 429
Response	HTTP 429, rate_limit_exceeded	High
Retry-after	0.165–9.818 s	High
Recovery	Not measured	—

3. Results

The CSV contains 57 requests: 22 real and 35 stand-in. The original real run had 15 requests; V2 added seven with TTFT instrumentation and a 500-token output limit.

Table 2a — Latency and offered load

Scenario	N	Duration	Req/min	TTFT p50/95/99	Total p50/95/99 ms
Original real	15	NM	NM	NM	1067/1451/1510
V2 real	7	NM	NM	765/1521/1526*	1102/1699/1702
Steady	5	1.96s	153	303/305/305	391/393/393
Burst	10	0.89s	671	801/802/802	885/889/889
Overload	20	2.81s	427	804/805/805	892/894/894
Halved limit / daily exhausted	—	NM	NM	NM	NM

*V2 TTFT has six observations; one request had no visible first token. Original latency uses six successful requests. NM = not measured; stand-in rates are approximate short-batch rates.

Table 2b — Outcomes

Scenario	Success	429	Timeout/error	Retries/req.	Queue	Tokens in/out	Cost/1k
Original real	40%	60%	0%	NM	NM	NM	NM
V2 real	100% API	0%	0%	NM	NM	1068/333 avg.	NM
Stand-in	100%	0%	0%	0	NM	NM	NM

V2: four correct, two incomplete/wrong and one correctly refused; two reached the 500-token cap. Across 13 manually graded answers: seven correct, four wrong/incomplete and two correctly refused. Q8's grade uses a later successful retest, not its original truncated output.

4. Where time goes

Original successful requests had total-latency p50 of 1.07 s; TTFT was not recorded. V2 TTFT p50 was 765 ms and total p50 1.10 s. Retrieval, queueing and inference were not measured separately.

5. When limit hit

Nine of 15 original requests received 429 responses identifying an 8,000 TPM limit. I stopped rather than exhaust quota. Retry tests used mocked 429s. V2 had no 429s, but two answers were incomplete at the output cap.

The Relay — Research Paper Assistant (continued)

Retrieval: quality versus cost

I compared TF-IDF with FastEmbed (BAAI/bge-small-en-v1.5) and reciprocal-rank fusion (RRF). Both retrieved the expected paper in the top three for all 12 in-scope questions (paper-level MRR 1.0). This does not prove passage relevance or answer correctness.

Table 3 — Retrieval settings

Setting	Chunk	Fetched	Rerank	Qs	Correct + cited	Wrong	Correct refusal	Tokens/answer	TTFT p50/95	Total p50/95
TF-IDF pipeline	220	3	No	15	NM	NM	NM	NM	NM	NM
TF-IDF evaluation	220	5	No	15	NM	NM	NM	NM	NM	NM
Hybrid	220	3	RRF	15	NM	NM	NM	NM	NM	NM

Answer generation used TF-IDF top three only; equivalent generation tests across settings were not run.

How this library will mislead a naive retriever

A Bengaluru population query retrieved causal-masking passages; an Olympic 100m query retrieved unrelated Moshi dialogue; a dot-product query returned a correct Attention passage alongside an unrelated learning-rate passage. A 0.10 threshold rejected relevant Q12 (0.0747) but accepted the unrelated Bengaluru result (0.1291). I removed the threshold because similarity is not answer confidence.

Your feedback on Vaani

The tool-using voice-agent workflow and webhook delivery stood out. The distinction between dispatch acceptance and call completion was useful. An end-to-end example showing a request, tool call and final webhook payload would improve clarity.

6. Retries

Two retries follow the initial attempt. Mock tests verified recovery after one 429 and stopping after three consecutive 429s. Partial streams are not retried.

7. What the person sees

Users receive streamed answers with source references. Unsupported questions receive: “The provided papers do not contain enough information to answer this question.” Irrelevant retrieval and token limits can still produce incomplete answers.

8. What I'm unsure about, what I left out, what I learned

RPM, daily limits, concurrency, queue length, cost and recovery time were not measured. Halved-limit and daily-exhaustion tests were not run. The stand-in does not reproduce model quality or Groq TPM accounting. I learned to measure retrieval relevance, answer correctness and service reliability separately.

How I worked with the assistant

Three reconstructed prompts: debug the async Groq client without rewriting the project; design stand-in tests; interpret raw results without inventing measurements. I rejected a fixed relevance threshold after the Bengaluru example and kept TF-IDF because hybrid showed no paper-level improvement. I implemented the pipeline, ran experiments, graded answers and made the final design choices; the assistant helped debug and review evidence.